SMART MOTORCYCLE SAFETY

RideGuardian AI

One tap starts a connected safety companion that monitors the ride, responds to possible danger, finds medical care, and keeps approved family members informed.

FLUTTER ANDROID APP / SERVERLESS AWS BACKEND

Client overview - implemented product, user journey, architecture, security, delivery status, and production roadmap.



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The product in one minute

A simple safety experience for the rider and clear, permission-controlled reassurance for the family.

For the rider

One button starts GPS, speed, motion, impact, route, scoring, and cloud monitoring. The same button ends the ride and stops continuous services.

For the family

Only accepted guardians can retrieve a shared rider's state, speed, score, last update, authorized location, and emergency alerts.

The rider journey

1	2	3	4	5
Verify account	Start ride	Monitor safety	Respond if needed	Save history

What makes it easy to understand

- **One persistent control:** no separate switches for GPS, sensors, cloud tracking, and ride scoring.
- **High-visibility states:** green means active/safe; red marks stop and emergency actions.
- **Plain-language feedback:** riding, safe, offline, alert sent, and system status are visible.
- **Privacy by invitation:** family sharing is not automatic and can only follow an accepted relationship.

Screenshot note. All frames in this document are real rendered Flutter widgets. Names, locations, and ride numbers are representative presentation data, not customer records.

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Secure access

Individual, verified accounts protect both rider and guardian experiences.

What the client sees

A focused account screen with email, password, account creation, and a clear AWS Cognito trust message.

- Email registration and verification code
- Secure sign-in and sign-out
- Automatic token refresh
- Separate accounts for riders and guardians

Why it matters

There are no AWS access keys or user passwords compiled into the application. Private API requests carry a short-lived Cognito JWT.

Privacy starts here

A guardian must sign in, receive an invitation, and accept it before rider status can be requested.



Secure Cognito sign-in frame

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One control for the whole ride

The persistent Start/End button is the operational center of the application.

Starting a ride activates

- Precise GPS and live location
- Current speed, maximum speed, and distance
- Accelerometer and gyroscope monitoring
- Harsh movement and possible-impact rules
- Family location heartbeats
- Ride scoring and history collection

Ending a ride

The same button stops sensor subscriptions and cloud heartbeats, publishes a final SAFE state, and saves the ride summary.

Resource-aware by design

The continuous phone services are active only during a ride. The AWS backend is serverless and performs work only when requests arrive.



Active-ride dashboard with master control

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Live ride and medical routing

The rider gets essential navigation and safety information without visual clutter.

Live ride information

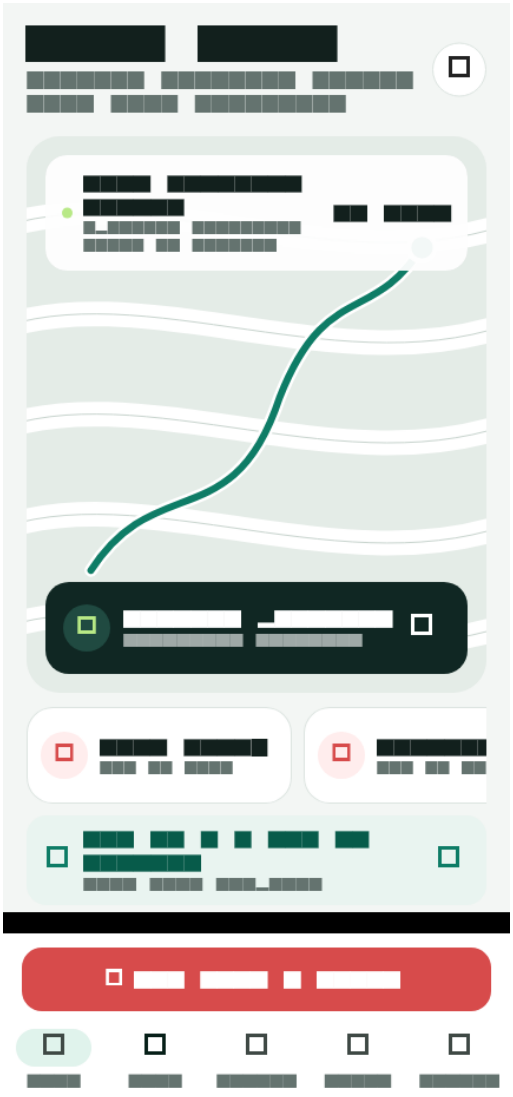
- Tracking state and current speed
- Current GPS coordinates
- Nearby medical facilities
- Selected hospital distance and estimated time

AWS-powered search

Amazon Location performs the real nearby-place search and scooter-route calculation. The selected location can be handed to Google Maps for navigation.

Private by default

The rider's live coordinates are sent through an authenticated API and are exposed only to accepted guardians.



Live tracking, nearby hospitals, and route estimate

RIDEGUARDIAN AI / 05

Safety center

Protection tools and system health are visible in one place.

Protection tools

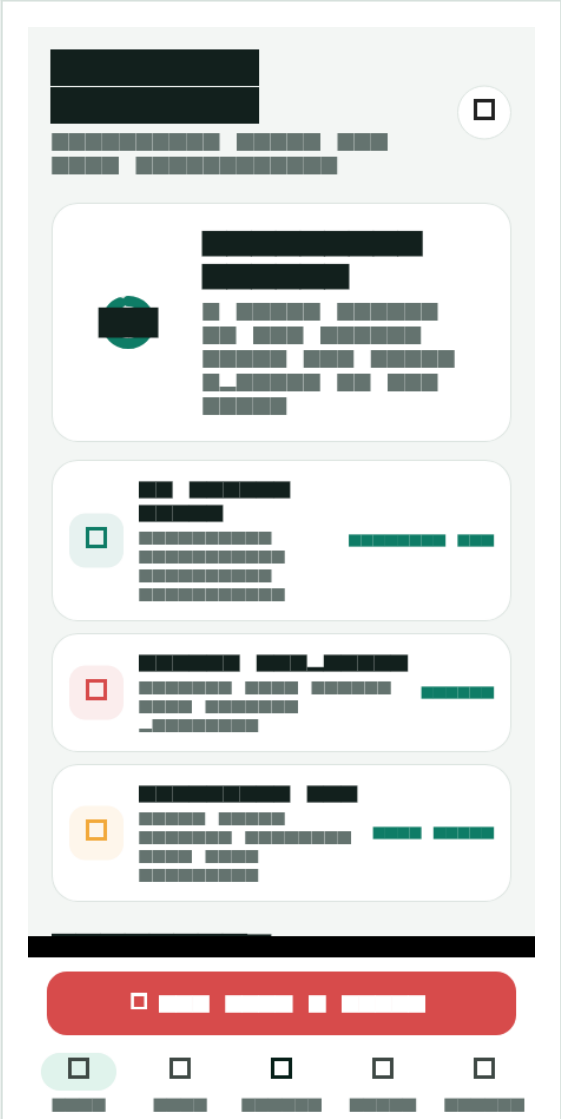
- Safety score and detected-risk count
- Pretrained AI helmet assessment
- Nearby hospital search
- Manual emergency SOS

System status

The app clearly shows whether speed monitoring, impact detection, rider behavior analysis, and family updates are active or standing by.

No custom model training

Helmet assessment calls the pretrained AWS Rekognition protective-equipment capability. The MVP does not require a training dataset.



Safety score, AI check, hospital search, and SOS

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Possible-accident response

A high-visibility countdown balances urgent response with false-alarm control.

How it responds

- A strong sensor pattern is marked as a possible impact
- The screen opens a 20-second cancellation window
- A safe rider can cancel the alert immediately
- If not cancelled, the latest location and reason are published
- Manual SOS remains available at any time

Why the countdown exists

Phone motion can be caused by drops, potholes, or sudden handling. The countdown gives a conscious rider a clear way to prevent an unnecessary emergency.

Safety MVP

Impact thresholds must be calibrated through controlled, consented road testing before production use.



Possible-impact countdown with cancel action

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Understand every ride

Cloud history turns safety activity into a progress record.

Each ride summary includes

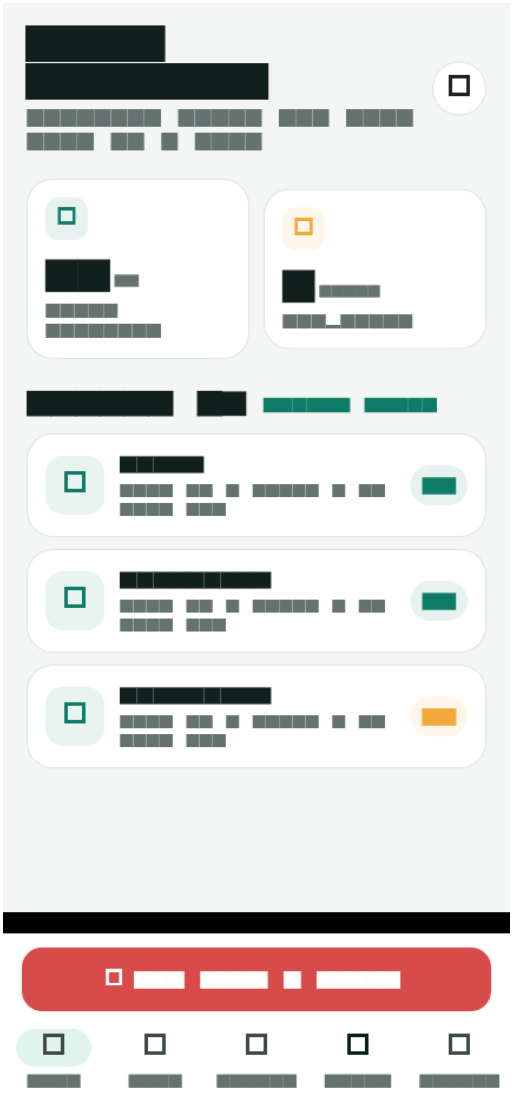
- Ride date and start time
- Distance and duration
- Maximum speed
- Detected event count
- Final safety score

Client value

The rider can understand trends rather than viewing safety as a one-time warning. History is private to the signed-in account and stored in DynamoDB.

Explainable score

The 0-100 score uses deterministic penalties for overspeed, harsh braking, acceleration, sharp turns, and possible impacts.



Private cloud ride-history summaries

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Family guardian circle

The rider decides who can see their safety status.

Invitation workflow

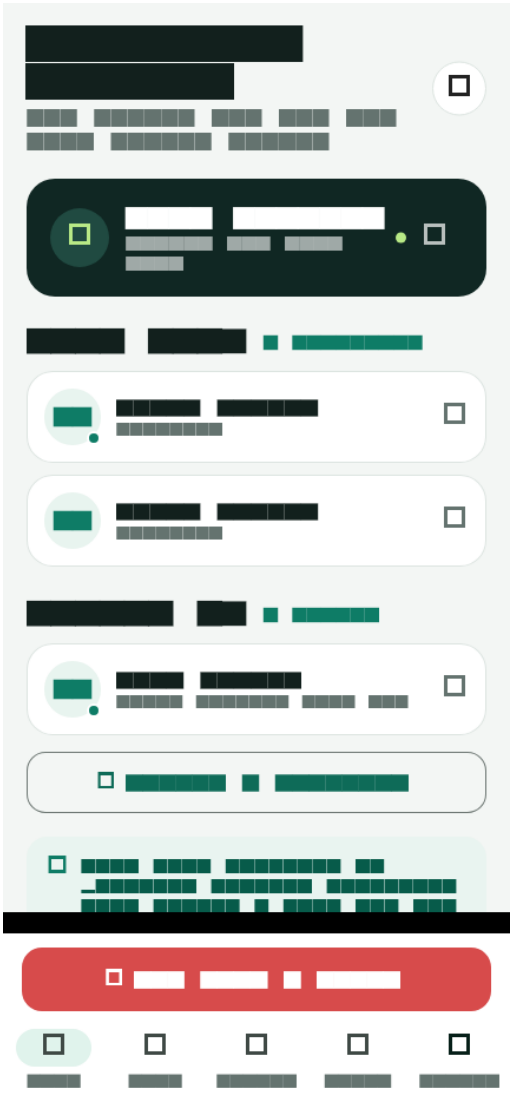
- Rider invites the guardian's exact account email
- Guardian signs in and accepts
- Backend creates the reciprocal relationship
- Every live-status request checks permission again

What an approved guardian can see

- Riding, safe, or offline state
- Safety score, speed, and last update
- Authorized last location
- In-app emergency alerts

Consent is explicit

Creating a family account alone does not reveal a rider. An accepted invitation is required.



Permission-controlled guardian and shared-rider view

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Implemented capability matrix

The current MVP connects the visible client experience to working mobile and AWS components.

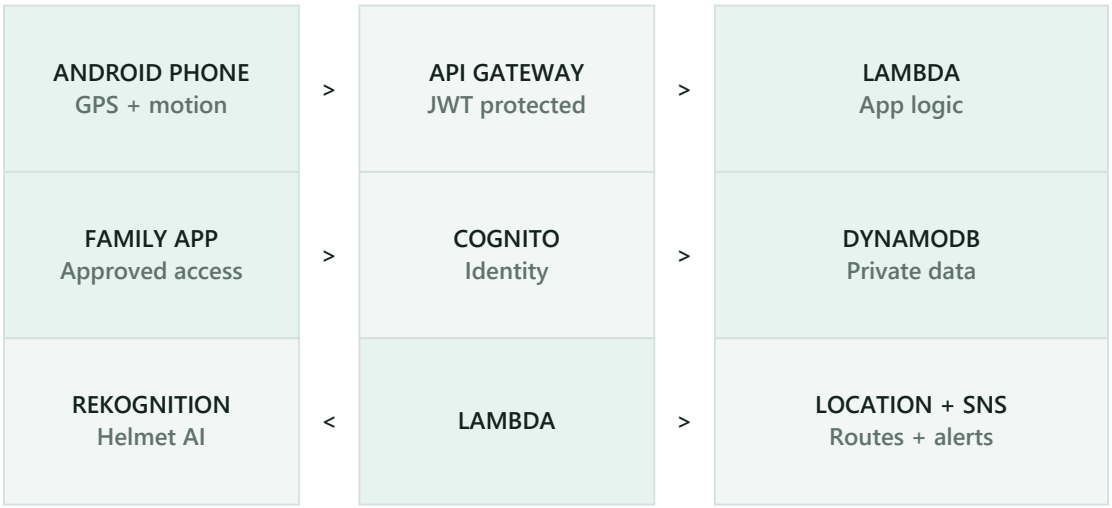
Capability	Client value	Implementation
Account security	Verified registration, login, logout	Cognito + Amplify
Universal ride control	One persistent Start/End button	Flutter lifecycle controller
Speed monitoring	Live and maximum km/h	Android GPS
Location tracking	Coordinates and cloud heartbeats	GPS + API + DynamoDB
Ride distance	Distance along route points	On-device calculation
Rider behavior	Harsh braking, acceleration, turns	Accelerometer + gyroscope
Accident assistance	Countdown and automatic SOS	Motion rules + Lambda + SNS
Manual SOS	Rider-triggered location alert	Authenticated emergency API
Helmet check	Photo confidence assessment	Pretrained Rekognition PPE
Nearby hospitals	Medical places near GPS fix	Amazon Location Places
Route estimate	Scooter distance and duration	Amazon Location Routes
Safety score	Clear 0-100 result	Deterministic scoring
Ride history	Private completed-ride summaries	DynamoDB
Family sharing	Accepted rider/guardian relationship	Cognito + DynamoDB
Family status	State, speed, score, time, location	Permission-checked API
Resource control	Sensors and heartbeats stop on End	Central ride lifecycle

Key distinction: AI is used where a pretrained visual model adds value. Core ride monitoring remains deterministic, understandable, and responsive on the phone.

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AWS architecture and security

A serverless design keeps operations simple and avoids a permanently running application server.



AWS service	Responsibility
Cognito User Pool	Verified accounts and secure tokens
API Gateway	Public health route and JWT-protected private routes
Lambda - Python 3.13 ARM64	Application logic and permission checks
DynamoDB	Live state, rides, invitations, relationships, alerts
Rekognition	Pretrained protective-equipment assessment
Amazon Location	Nearby hospitals and scooter routes
SNS	Encrypted emergency-alert topic
CloudFormation / SAM	Repeatable deployment

Backend safeguards

- JWT authentication on every private route
- Permission checks for guardian access
- No AWS keys or passwords inside the APK
- DynamoDB encryption, point-in-time recovery, and retained data protection
- Encrypted SNS topic and scoped Lambda permissions
- Pay-per-request storage and request-driven compute

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How to demonstrate the product

A two-account walkthrough makes both sides of the safety experience easy for a client to understand.

Rider device

- Create and verify the rider account
- Allow location, camera, motion, and notification permissions
- Press Start ride safety systems
- Show Live, Safety, and Ride history
- Invite the family member's exact email
- Press End ride to stop services and save the summary

Family device

- Create and verify a second account
- Open Family and refresh invitations
- Accept the rider's invitation
- Open the shared rider status
- Show state, speed, score, update time, and location
- Refresh the Family view to retrieve alerts

Current delivery status

AWS backend	Deployed and healthy - UPDATE_COMPLETE
API health	HTTP 200
Flutter analyzer	No issues
Widget tests	Passing
Screenshot render test	Passing at 430 x 932
Android debug APK	Built successfully

APK: build/app/outputs/flutter-apk/app-debug.apk
SHA-256: 106E963A23C5118725CCC39C58D133027C01A43E9FE17C5E1CFEFDA20EF29755

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Production roadmap

The MVP is working, but a safety product requires careful validation and operational preparation before public release.

1. Controlled road calibration	Calibrate impact and behavior thresholds across different phones, mounting positions, motorcycles, and road conditions.
2. Push and SMS delivery	Connect Firebase Cloud Messaging and/or an approved SMS channel. The current Family view retrieves stored alerts; SNS is provisioned without customer phone subscriptions.
3. Privacy lifecycle	Add explicit consent, retention settings, guardian removal, data export, and account deletion.
4. Store-policy readiness	Complete Google Play background-location disclosure, policy review, release signing, and permission education.
5. Device validation	Add physical-device integration tests, accessibility checks, battery profiling, network-loss scenarios, and emergency drills.
6. Production operations	Add alarms, dashboards, audit trails, abuse controls, throttling, delivery monitoring, and incident response.

Important: RideGuardian is a working safety MVP, not a certified crash-detection or medical device. It must not replace responsible riding, protective equipment, or local emergency services.

One tap for a safer ride and a better-informed family.

RideGuardian AI combines a clear mobile experience with authenticated, serverless AWS services to create a credible foundation for the next stage of product validation.